



Closing Remarks & Way Forward

NATO Science for Peace and Security Advanced Study Institute (ASI)

Advanced Study Institute: Artificial Intelligence for Disaster Management
Orlando, November 17 – 25, 2025



*This activity
is supported by:*

The NATO Science for Peace
and Security Programme

The Learning Path: From Pixels to Decisions

ARTIFICIAL INTELLIGENCE FOR DISASTER MANAGEMENT

Building Damage Identification and Flood
Event Detection Using Deep Learning



Days 2-3 (Foundations): Understanding the "Eye in the Sky" (Satellite/Aerial imagery) and the "Brain" (CNNs).

Days 4-5 (Applications): Building damage assessment and flood mapping (The "What" and "Where").

Day 6 (Integration): End-to-end workflows.

Days 7 (Deployment): Edge computing, optimization, and making models work in the mud.

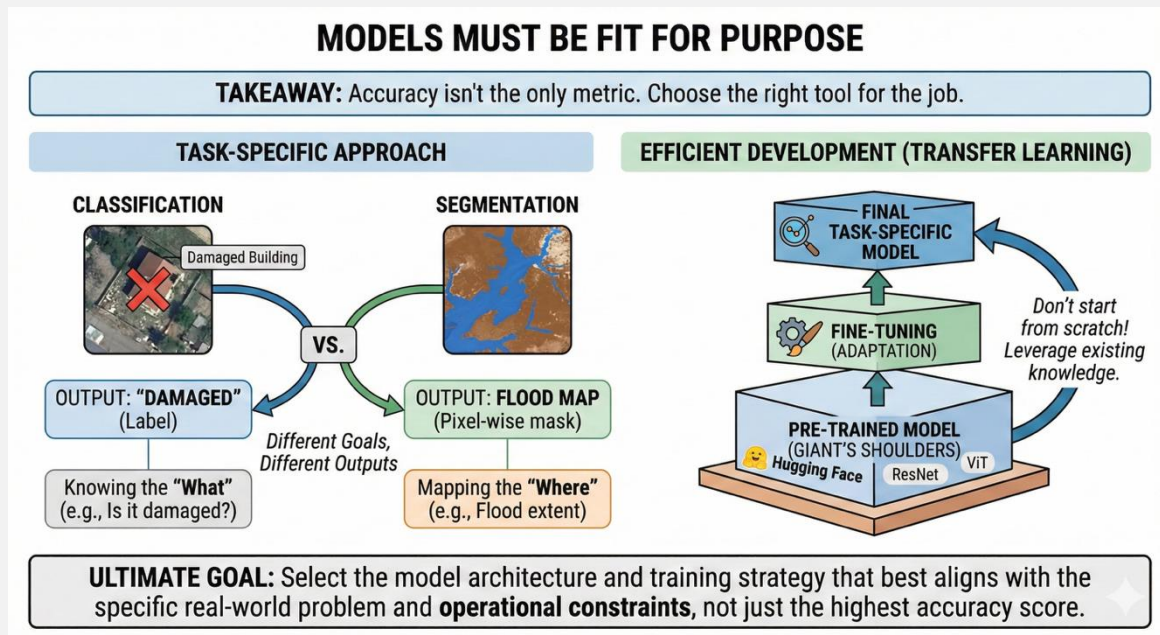
Key Takeaways: Data is the Hardest Part



Takeaway: We learned that real-world disaster data is messy, noisy, and unbalance.

Key Concepts: Data augmentation, the importance of labeling (xView2, BigEarthNet), and handling "DDIL" (Denied, Degraded, Intermittent, Limited) environments.

Key Takeaways: Models Must Be Fit for Purpose



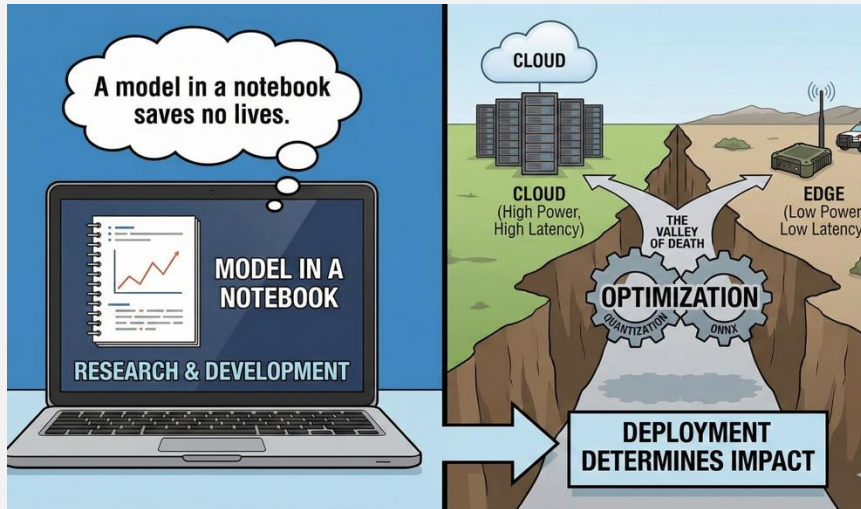
Takeaway: Accuracy isn't the only metric.

Key Concepts:

Classification vs. Segmentation (knowing a building is damaged vs. mapping the flood extent).

Transfer Learning: Don't start from scratch; stand on the shoulders of giants (Hugging Face, ResNet, ViT).

Key Takeaways: Deployment Determines Impact

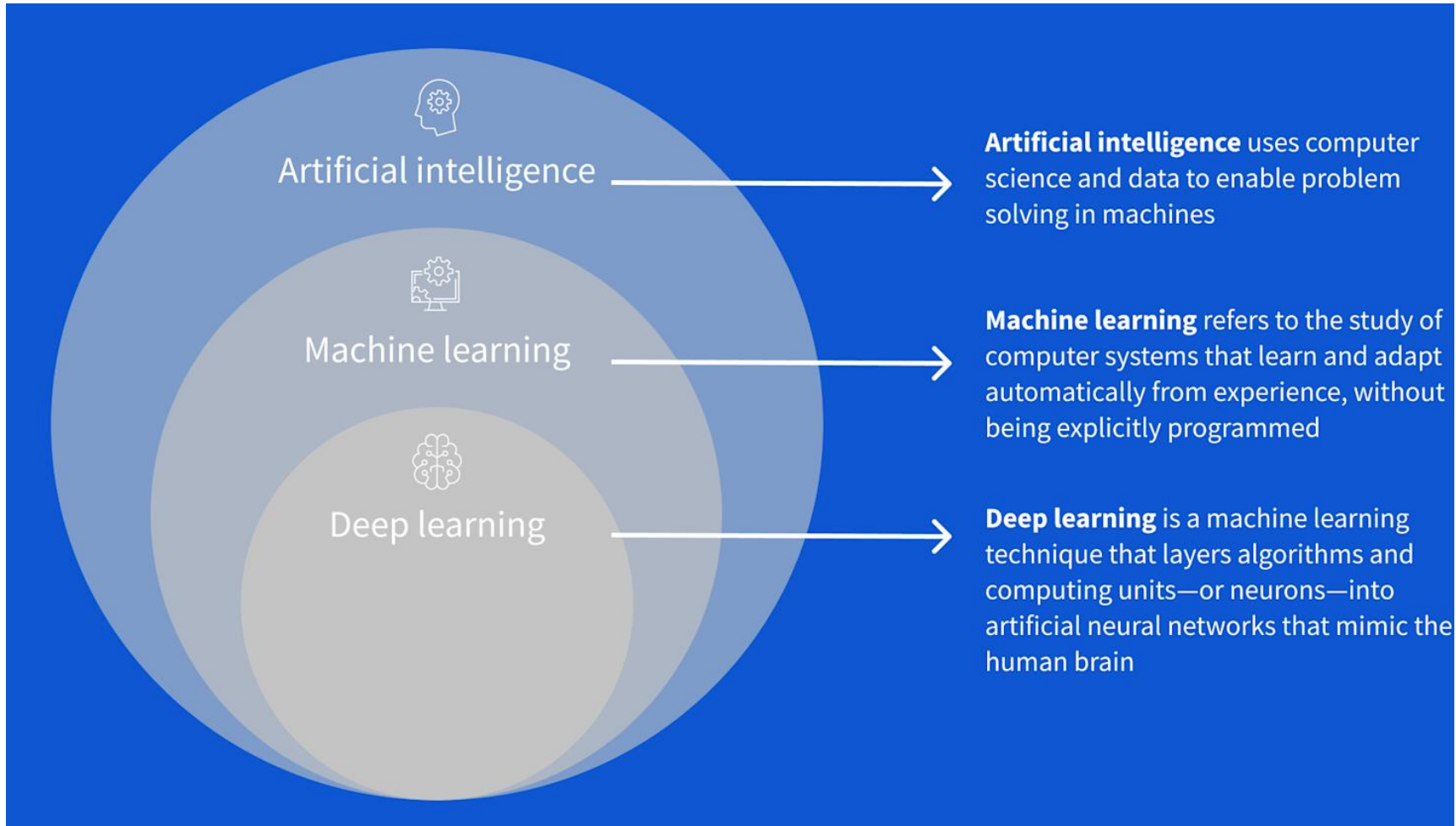


Takeaway: A model in a notebook saves no lives.

Key Concepts:

- The "Valley of Death."
- Edge vs. Cloud: The trade-off between latency and power.
- Optimization: Quantization and ONNX as essential tools for fielding AI.

Artificial Intelligence



Goals of This Advanced Study Institute



Knowledge Transfer

To provide a curriculum on the state-of-the-art in AI and DL for disaster management



Practical Application

To move beyond theory with hands-on training, enabling you to apply these advanced techniques to real-world problems



Collaboration & Networking

To build a lasting, international network of next-generation researchers and practitioners in this critical field



Addressing a Critical Challenge

To generate new insights and strategies for identifying building damage and flood events, ultimately saving lives and improving resilience

The NATO Science Series Book



Action Item: This is the major deliverable.

Details:

- All speakers and participants are contributors.
- **Timeline:**
 - Abstracts/Drafts due: Jan 2026
 - Review Process: March 2026
 - Final Camera-Ready Submission: August 2026 (9 months post-event).
- **Format:** Open Access, fully funded by the grant.

Acknowledgments: Thank You to the Team

- **Sponsor:** NATO Science for Peace and Security (SPS) Programme.
- **Co-Director:** Distinguished Prof. Saeid Nahavandi (Swinburne University).
- **Organizing Committee:** Dr. Ghaith Rabadi, Dr. Soheil Sabri.
- **The Student Support Team:**
 - Costi & Jedidah (Logistics/Registration).
 - Safoora & Sherry (Virtual Moderation).
- **Instructors & Speakers:** For sharing their expertise.

Final Call to Action

The technology we discussed is powerful, but the network we built this week is stronger.

Don't let the conversation end here. Collaborate on papers, apply for joint grants, and keep solving these critical problems.



Thank you for your attention!

Q&A